

6	(a)	Characteristic lines When an <u>inner</u> shell electron is knocked out by a bombarding electron and an outer shell electron fills in this vacancy, this transition results in X-ray photons of a specific frequency to be emitted. Continuous spectrum When the electrons are <u>accelerating towards the target and when they are decelerating when hitting the target, electromagnetic waves will be emitted</u>. In this case, the EM waves emitted is of high energy and is in the X-rays region. Because the acceleration and deceleration are <u>continuous processes, the energy converted to X-rays can be any values</u>. Hence continuous spectrum is formed.		B1 B2 B3
	(b)	(i)	From the graph, $\lambda_{min} = 0.8 \times 10^{-11} m$ Max energy of x-ray photons $E_{k,max} = \frac{hc}{\lambda_{min}} = 2.49 \times 10^{-14} J$ [ECF for λ_{min}] Note: KE of accelerated electron is entirely converted into an X-ray photon in a (single) collision betw the electron & the target metal.	B1 A1
		(ii)	$e V_a = \text{Max energy of the x-ray photon} = 2.49 \times 10^{-14} J$ accelerating pd $V_a = 2.49 \times 10^{-14} J \div e = 155 kV$	B1
		(iii)	$qV = \frac{1}{2}mv^2 = \text{Max energy of x-ray photons from (b)(i)}$ $v = \sqrt{\frac{2qV}{m}}$ $= \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 155\,000}{9.11 \times 10^{-31}}} = 2.33 \times 10^{-12} m s^{-1}$	B1 A1
	(c)		$\lambda = \frac{h}{p} = \frac{h}{mv}$ $= \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 2.33 \times 10^8} = 3.12 \times 10^{-12} m$ The wavelength of the electrons is much smaller than the atomic spacing of the metal target. Hence, diffraction effects {a wave property} are not observable. {The atomic planes act like a diffraction grating; for significant diffraction, λ must $\approx$ atomic spacing, about $10^{-10} m$} Note: the radius of proton or neutron is $\sim 10^{-15} m$	A1 B1